Beaches_Bikes_RSM

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This notebook was produced to aid RSM students with some data analysis for their project. They mainly wanted data on Daily breakdown (weekday vs. weekend), Weekly averages, and Full-year data to assess seasonality trends.

Importing Data and Libraries

```r
library(opendatatoronto)
```

```
Warning: package 'opendatatoronto' was built under R version 4.5.2
```

```r
library(tidyverse)
```

```
Warning: package 'purrr' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(sf)
```

```
Warning: package 'sf' was built under R version 4.5.2
```

```
Linking to GEOS 3.13.1, GDAL 3.11.4, PROJ 9.7.0; sf_use_s2() is TRUE
```

```r
library(leaflet)
```

```
Warning: package 'leaflet' was built under R version 4.5.2
```

```r
library(skimr)
```

```
Warning: package 'skimr' was built under R version 4.5.2
```

```r
library(janitor)
```

```
Warning: package 'janitor' was built under R version 4.5.2


Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(lubridate)
library(tidytext)
```

```
Warning: package 'tidytext' was built under R version 4.5.2
```

```r
set.seed(2453)
```

```r
df <- read.csv("../01_Data/beaches_bikes.csv")
```

Analysis for RSM Students

```r
df_clean <- df %>%
  mutate(
    started_at = ymd_hms(datetime_start),
    date = as.Date(started_at),
    # We recreate the year and month safely from the date object for correct sorting
    year = year(date),
    month = month(date, label = TRUE, abbr = TRUE),
    # Create Day of Week and Weekend/Weekday flag
    day_of_week = wday(date, label = TRUE, abbr = FALSE),
    day_type = ifelse(day_of_week %in% c("Saturday", "Sunday"), "Weekend", "Weekday"),
    # Create a continuous weekly identifier for multi-year trending
    year_week = floor_date(date, "week")
  ) %>%
  # Filter out any rows with missing dates that might break the plots
  filter(!is.na(date))

head(df_clean)
```

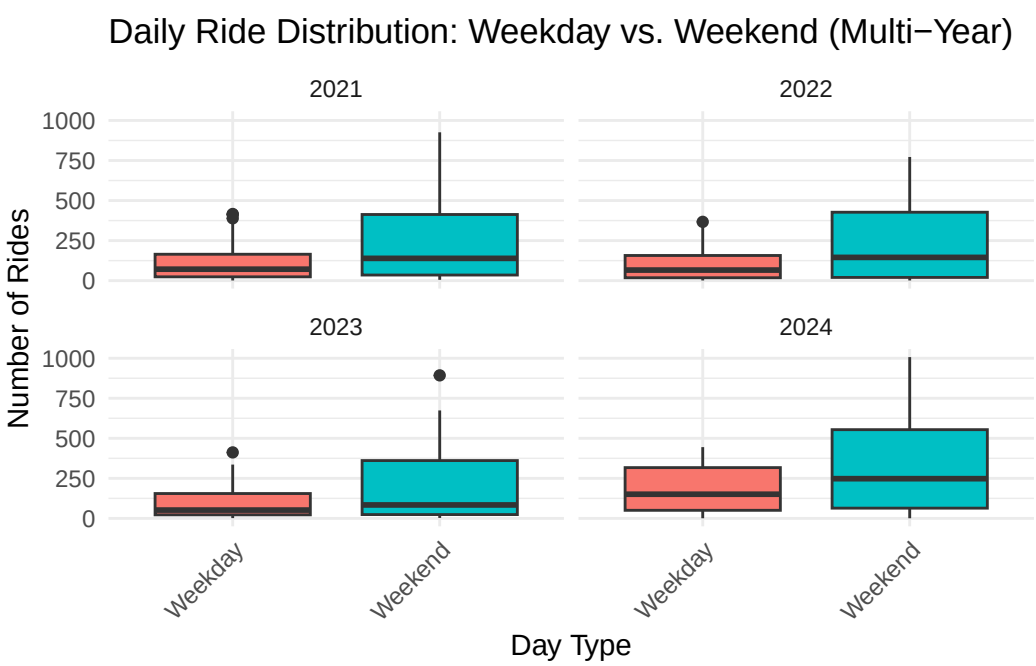
```
   Trip.Id Trip..Duration Start.Station.Id       Start.Time
1 10644802            352             7446 01/01/2021 11:12
2 10644809            570             7427 01/01/2021 11:18
3 10644836            962             7318 01/01/2021 11:28
4 10645084            997             7644 01/01/2021 13:05
5 10645129            461             7427 01/01/2021 13:23
6 10645153            961             7429 01/01/2021 13:29
                        Start.Station.Name End.Station.Id         End.Time
1                             Orchard Park           7427 01/01/2021 11:18
2 Northern Dancer Blvd / Lake Shore Blvd E           7318 01/01/2021 11:28
3           Hubbard Blvd. / Glen Manor Dr.           7354 01/01/2021 11:44
4             East Lynn Ave / Danforth Ave           7427 01/01/2021 13:21
5 Northern Dancer Blvd / Lake Shore Blvd E           7330 01/01/2021 13:30
6          Woodbine Subway Station - SMART           7427 01/01/2021 13:45
                          End.Station.Name Bike.Id     User.Type ï..Trip.Id
1 Northern Dancer Blvd / Lake Shore Blvd E    5461 Annual Member         NA
2           Hubbard Blvd. / Glen Manor Dr.    5461 Annual Member         NA
3 Tommy Thompson Park (Leslie Street Spit)    5461 Annual Member         NA
4 Northern Dancer Blvd / Lake Shore Blvd E    3347 Casual Member         NA
5                     Lee Ave / Queen St E    3347 Casual Member         NA
6 Northern Dancer Blvd / Lake Shore Blvd E    2337 Annual Member         NA
  Model       datetime_start start_date_actual start_hour_of_day
1  <NA> 2021-01-01T11:12:00Z        2021-01-01                11
2  <NA> 2021-01-01T11:18:00Z        2021-01-01                11
```

```
3  <NA> 2021-01-01T11:28:00Z        2021-01-01                11
4  <NA> 2021-01-01T13:05:00Z        2021-01-01                13
5  <NA> 2021-01-01T13:23:00Z        2021-01-01                13
6  <NA> 2021-01-01T13:29:00Z        2021-01-01                13
  start_month_name year         datetime_end end_date_actual end_hour_of_day
1              Jan 2021 2021-01-01T11:18:00Z      2021-01-01              11
2              Jan 2021 2021-01-01T11:28:00Z      2021-01-01              11
3              Jan 2021 2021-01-01T11:44:00Z      2021-01-01              11
4              Jan 2021 2021-01-01T13:21:00Z      2021-01-01              13
5              Jan 2021 2021-01-01T13:30:00Z      2021-01-01              13
6              Jan 2021 2021-01-01T13:45:00Z      2021-01-01              13
  end_month_name          started_at       date month day_of_week day_type
1            Jan 2021-01-01 11:12:00 2021-01-01   Jan      Friday  Weekday
2            Jan 2021-01-01 11:18:00 2021-01-01   Jan      Friday  Weekday
3            Jan 2021-01-01 11:28:00 2021-01-01   Jan      Friday  Weekday
4            Jan 2021-01-01 13:05:00 2021-01-01   Jan      Friday  Weekday
5            Jan 2021-01-01 13:23:00 2021-01-01   Jan      Friday  Weekday
6            Jan 2021-01-01 13:29:00 2021-01-01   Jan      Friday  Weekday
   year_week
1 2020-12-27
2 2020-12-27
3 2020-12-27
4 2020-12-27
5 2020-12-27
6 2020-12-27
```

```r
#  Define outlier function
is_outlier <- function(x) {
  q1 <- quantile(x, 0.25, na.rm = TRUE)
  q3 <- quantile(x, 0.75, na.rm = TRUE)
  iqr <- q3 - q1
  return(x < (q1 - 1.5 * iqr) | x > (q3 + 1.5 * iqr))
}
```

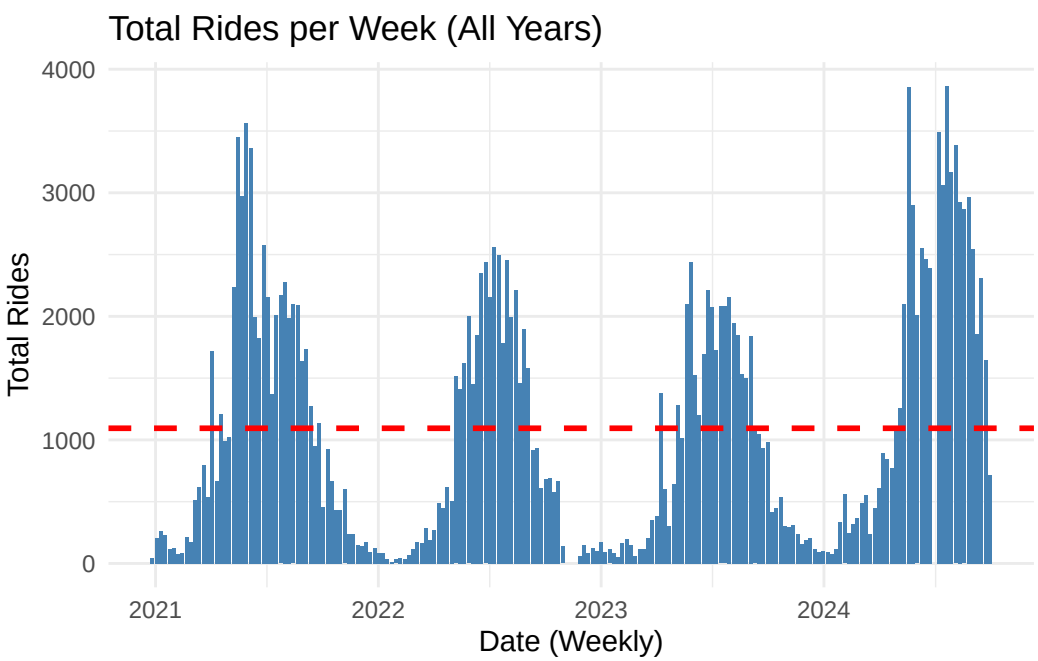
```r
# Daily Breakdown
daily_counts_clean <- df_clean %>%
  group_by(date, day_type, year) %>%
  summarise(total_rides = n(), .groups = 'drop') %>%
  group_by(day_type) %>%
  filter(!is_outlier(total_rides)) %>%
  ungroup()
```

```r
# plot the boxplots faceted by year
ggplot(daily_counts_clean, aes(x = day_type, y = total_rides, fill = day_type)) +
  geom_boxplot() +
  facet_wrap(~ year) +
  labs(title = "Daily Ride Distribution: Weekday vs. Weekend (Multi-Year)",
       y = "Number of Rides",
       x = "Day Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```

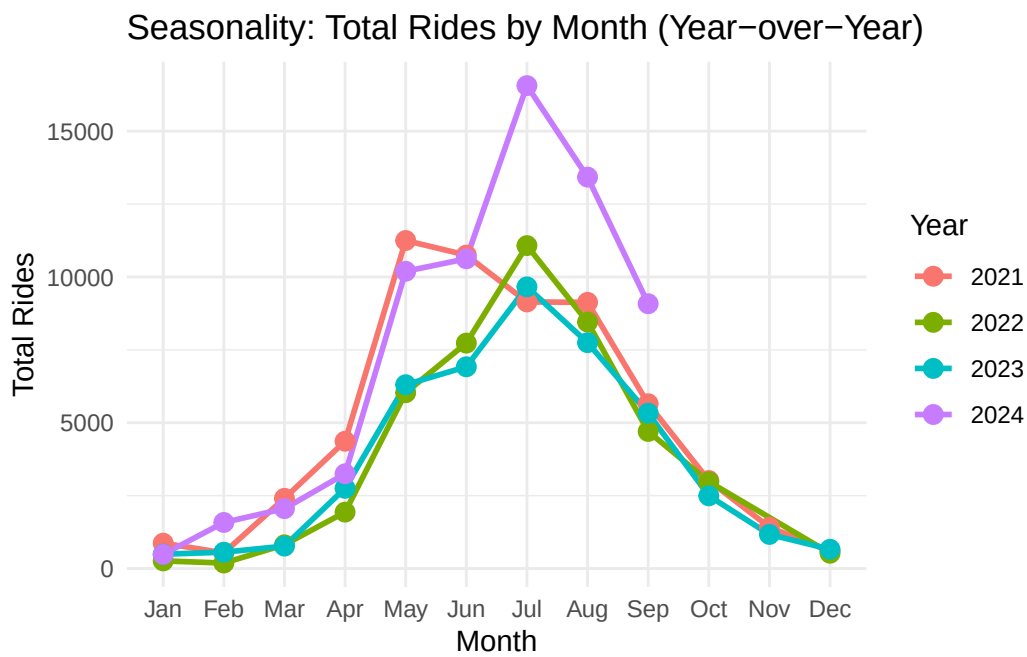



```r
# Weekly Averages
weekly_data_clean <- df_clean %>%
  group_by(year_week) %>%
  summarise(total_weekly_rides = n(), .groups = 'drop') %>%
  filter(!is_outlier(total_weekly_rides))

overall_weekly_avg <- mean(weekly_data_clean$total_weekly_rides)

# plot chart over all years
ggplot(weekly_data_clean, aes(x = year_week, y = total_weekly_rides)) +
  geom_col(fill = "steelblue") +
```

```
  geom_hline(yintercept = overall_weekly_avg, linetype = "dashed", color = "red", linewidth =
  labs(title = "Total Rides per Week (All Years)",
       x = "Date (Weekly)",
       y = "Total Rides") +
  theme_minimal()
```

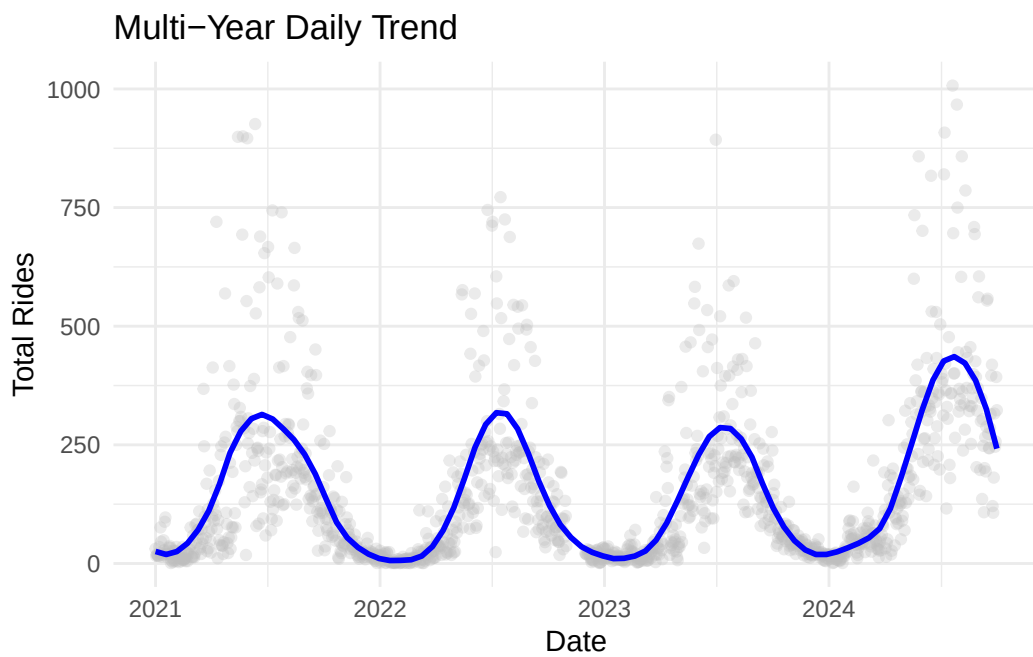



```
# seasonality trends
monthly_trend_clean <- df_clean %>%
  group_by(year, month) %>%
  summarise(total_rides = n(), .groups = 'drop') %>%
  filter(!is_outlier(total_rides))

# plot
ggplot(monthly_trend_clean, aes(x = month, y = total_rides, group = year, color = as.factor(y
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(title = "Seasonality: Total Rides by Month (Year-over-Year)",
       x = "Month",
       y = "Total Rides",
       color = "Year") +
  theme_minimal()
```




```r
# multi year daily trend with smoothing
ggplot(daily_counts_clean, aes(x = date, y = total_rides)) +
  geom_point(alpha = 0.3, color = "gray") +
  geom_smooth(method = "loess", color = "blue", se = FALSE, span = 0.15) +
  labs(title = "Multi-Year Daily Trend",
       x = "Date",
       y = "Total Rides") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
# Exporting the general analysis data
write.csv(daily_counts_clean, "daily_ride_distribution.csv", row.names = FALSE)
write.csv(weekly_data_clean, "weekly_ride_averages.csv", row.names = FALSE)
write.csv(monthly_trend_clean, "monthly_seasonality_trends.csv", row.names = FALSE)
```

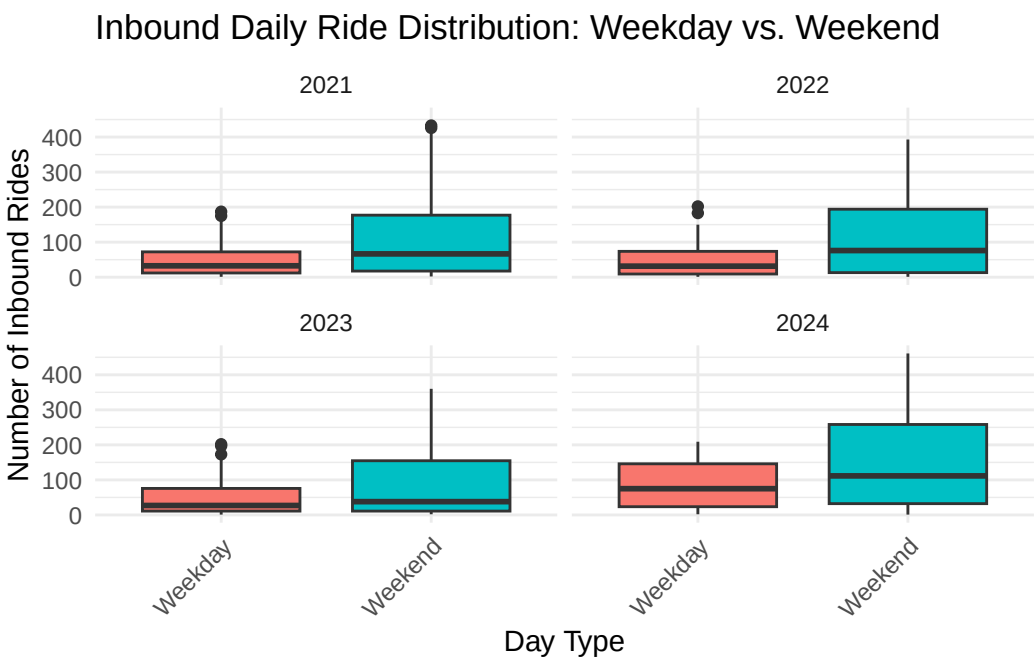
Repeat Analysis for Inbound Trips Only

```
# variable for beaches stations only
beaches_stations <- c(
  "7692",
  "7317",
  "7318",
  "7365",
  "7427",
  "7315",
  "7695",
  "7364",
  "7314",
  "7316",
  "7428"
)
```

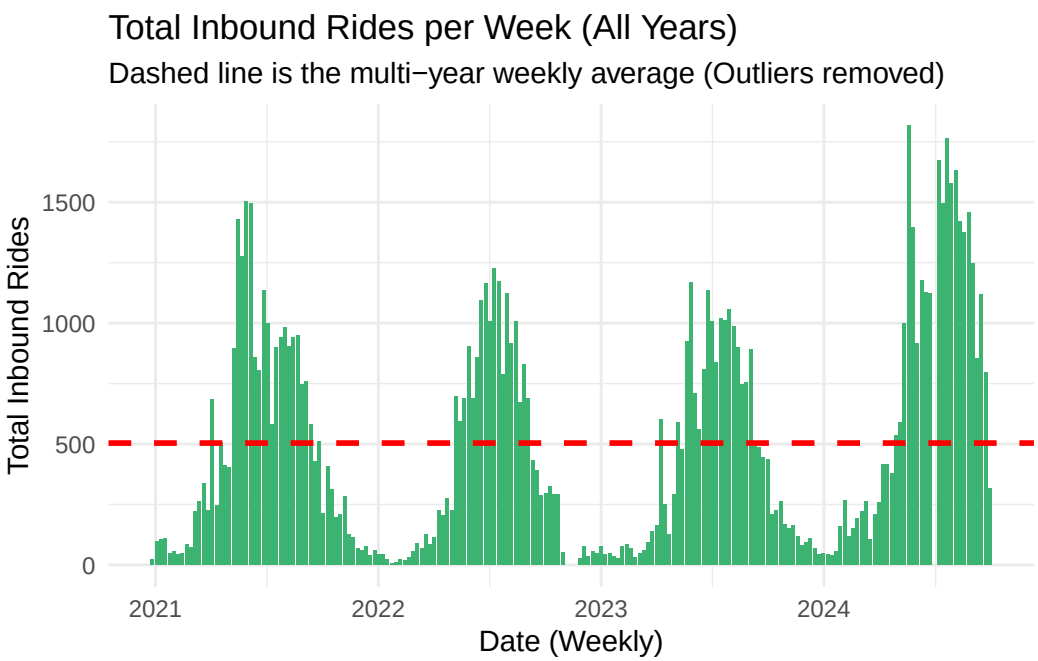
```r
# Create the inbound filtered dataset
df_inbound <- df %>%
  # trip ends in beaches
  filter(End.Station.Id %in% beaches_stations) %>%
  # trip starts outside of beaches
  filter(!(Start.Station.Id %in% beaches_stations))
```

```r
# preprocess data
df_clean <- df_inbound %>%
  mutate(

    started_at = ymd_hms(datetime_start),
    date = as.Date(started_at),

    year = year(date),
    month = month(date, label = TRUE, abbr = TRUE),

    day_of_week = wday(date, label = TRUE, abbr = FALSE),
    day_type = ifelse(day_of_week %in% c("Saturday", "Sunday"), "Weekend", "Weekday"),

    year_week = floor_date(date, "week")
  ) %>%
  filter(!is.na(date))
```

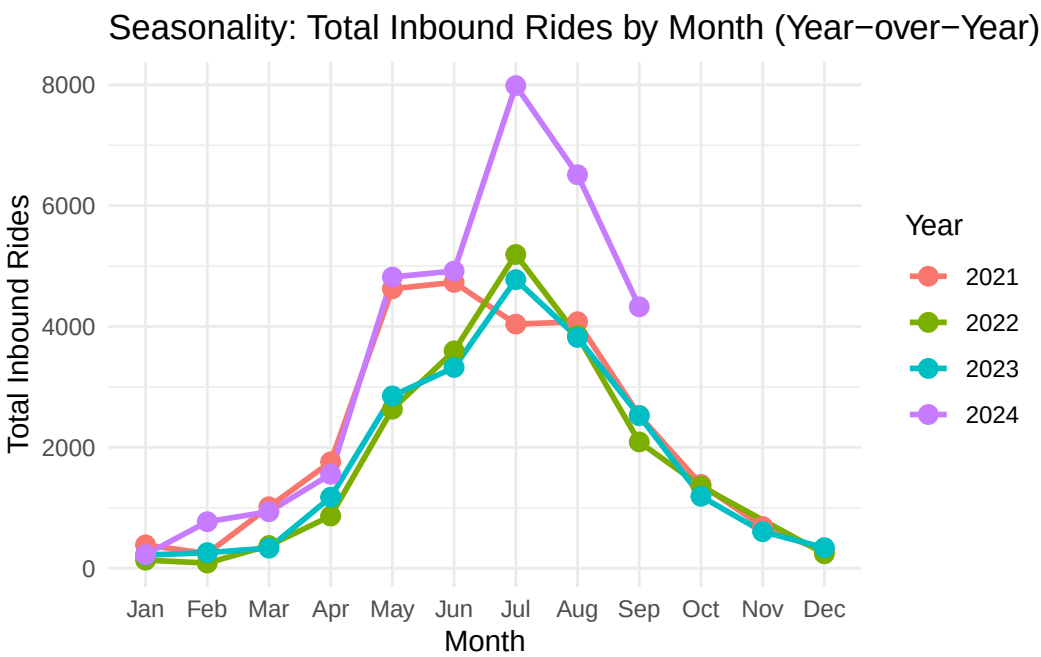
```r
# daily breakdown
daily_counts_clean <- df_clean %>%
  group_by(date, day_type, year) %>%
  summarise(total_rides = n(), .groups = 'drop') %>%
  group_by(day_type) %>%
  filter(!is_outlier(total_rides)) %>%
  ungroup()

ggplot(daily_counts_clean, aes(x = day_type, y = total_rides, fill = day_type)) +
  geom_boxplot() +
  facet_wrap(~ year) +
  labs(title = "Inbound Daily Ride Distribution: Weekday vs. Weekend",
       y = "Number of Inbound Rides",
       x = "Day Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```




```r
# weekly averages
weekly_data_clean <- df_clean %>%
  group_by(year_week) %>%
  summarise(total_weekly_rides = n(), .groups = 'drop') %>%
  filter(!is_outlier(total_weekly_rides))

overall_weekly_avg <- mean(weekly_data_clean$total_weekly_rides)

ggplot(weekly_data_clean, aes(x = year_week, y = total_weekly_rides)) +
  geom_col(fill = "mediumseagreen") +
  geom_hline(yintercept = overall_weekly_avg, linetype = "dashed", color = "red", linewidth =
  labs(title = "Total Inbound Rides per Week (All Years)",
       subtitle = "Dashed line is the multi-year weekly average (Outliers removed)",
       x = "Date (Weekly)",
       y = "Total Inbound Rides") +
  theme_minimal()
```

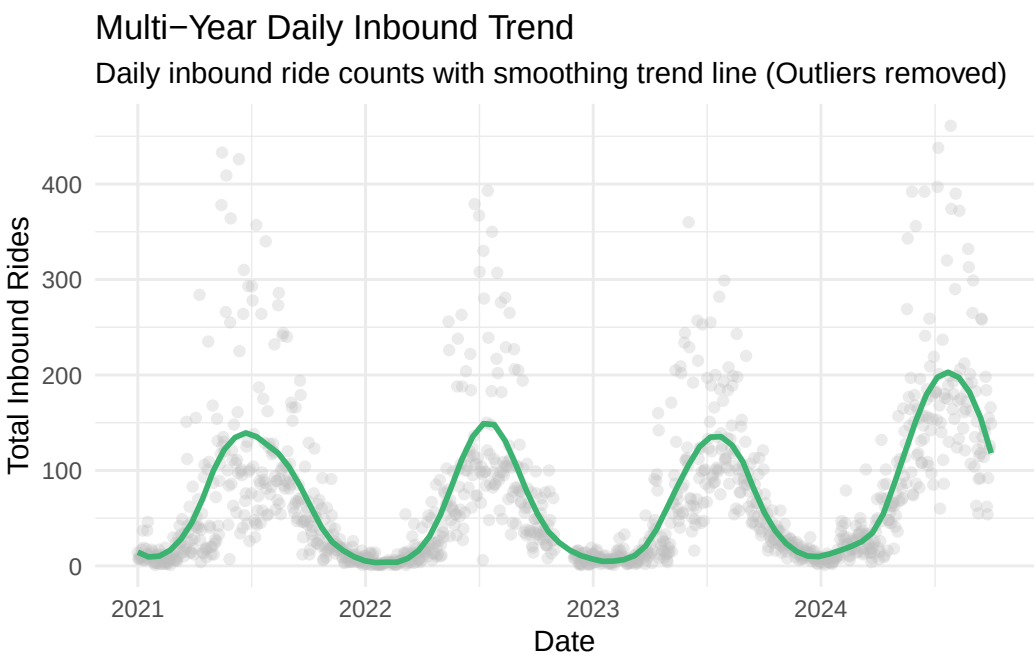



```
# seasonality trends
monthly_trend_clean <- df_clean %>%
  group_by(year, month) %>%
  summarise(total_rides = n(), .groups = 'drop') %>%
  filter(!is_outlier(total_rides))

ggplot(monthly_trend_clean, aes(x = month, y = total_rides, group = year, color = as.factor(y
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(title = "Seasonality: Total Inbound Rides by Month (Year-over-Year)",
       x = "Month",
       y = "Total Inbound Rides",
       color = "Year") +
  theme_minimal()
```




```
# multi year daily trend with smoothing
ggplot(daily_counts_clean, aes(x = date, y = total_rides)) +
  geom_point(alpha = 0.3, color = "gray") +
  geom_smooth(method = "loess", color = "mediumseagreen", se = FALSE, span = 0.15) +
  labs(title = "Multi-Year Daily Inbound Trend",
       subtitle = "Daily inbound ride counts with smoothing trend line (Outliers removed)",
       x = "Date",
       y = "Total Inbound Rides") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
# Exporting the general analysis data
write.csv(daily_counts_clean, "daily_ride_distribution_inbound.csv", row.names = FALSE)
write.csv(weekly_data_clean, "weekly_ride_averages_inbound.csv", row.names = FALSE)
write.csv(monthly_trend_clean, "monthly_seasonality_trends_inbound.csv", row.names = FALSE)
```